Jerome Pitogo de Leon, Ph.D.

Postdoctoral researcher at The University of Tokyo

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Overview

Data scientist and astrophysicist with 5+ years of experience building complex data analysis pipelines, statistical models, and scientific software. Proven track record of leading international research projects, analyzing large-scale time-series datasets from NASA's space telescopes, and delivering reproducible insights using modern probabilistic programming frameworks. Strong background in Bayesian inference, optimization, and mentoring cross-functional teams.

Skills

Programming & Scientific Computing: Python (NumPy, SciPy, Pandas, JAX, NumPyro); experience in building complex data analysis pipelines for large temporal and spatial datasets

Data Analysis & Modeling: Time-series analysis, Bayesian inference, probabilistic modeling, optimization, statistical analysis of large survey data, technical writing, project management

Software & Systems: Linux/Unix environments, server administration (Grafana monitoring), Bash scripting, version control (Git), automation, reproducible research workflows

Visualization & Communication: Matplotlib, scientific visualization, academic writing, conference presentations, public outreach

Leadership & Mentorship: Research group supervision, student advising (PhD, MSc, undergraduate), curriculum design, international collaboration

Languages: Filipino (native), English (fluent), Japanese (working proficiency)

Full-time Positions

Project Assistant Professor (2025/4 - Present)
Komaba Institute for Science, Graduate School of Arts and Sciences, The University of Tokyo, Japan
Role: Management of observations and data analysis with MuSCAT4 under the Kiban S project.

Project Research Assistant (2023/4 - 2025/3)
Department of Multi-Disciplinary Sciences, Graduate School of Arts and Sciences, The University of Tokyo, Japan
Role: Spearheading the discovery and in-depth characterization of young transiting exoplanets using the TESS telescope. Advisor: Prof. Norio Narita (Refer to de Leon et al. 2023)

Project Research Assistant (2021/4 - 2023/3)
Department of Astronomy, Graduate School of Science, The University of Tokyo, Japan
Role: Led an international group of astronomers towards the discovery of 37 new transiting exoplanets using the Kepler telescope. Advisor: Prof. Motohide Tamura (Refer to de Leon et al. 2021)

Research Development Assistant (2013/4-2014/6)
Regulus SpaceTech, Inc., Los Baños, Laguna, Philippines
Role: Initiated the software development for automating solar telescope control and data analysis. Advisor: Dr. Rogel Sese

Temporary Positions

Remote Lecturer (2021/6 - 2021/10)
Faculty of Aerospace Engineering, School of Engineering & Architecture, Ateneo de Davao University, Philippines

Role: Crafted and conducted lectures on Introduction to Astronomy & Astronomical Data Analysis.

Technical Assistant (2018/10-2019/3)
Engineering Department, The University of Tokyo, Japan
Role: Developed innovative image processing software for stitching featureless high-resolution images
Advisor: Assoc. Prof. Tomohiko Hayakawa (Refer to Murakami et al. 2018)

Research Assistant (2017/4-2018/3)
Department of Astronomy, The University of Tokyo, Japan
Role: Developed a robust pipeline for modeling and analyzing multi-wavelength transit light curves and provided mentorship to new research group members.

Research Student (2015/9-2016/3)
Department of Astronomy, Graduate University for Advanced Studies (SOKENDAI), Japan
Role: Worked on data reduction and analysis of high-resolution astronomical images and contributed to scientific report writing. Advisor: Dr. Satoshi Mayama

Intern (2014/7-2014/9)
Academia Sinica Institute for Astronomy and Astrophysics (ASIAA), Taipei, Taiwan
Role: Developed an image processing pipeline for the analysis of high-resolution images of a young forming star using the 8-m Subaru telescope. Advisor: Dr. Michihiro Takami (Refer to de Leon et al. 2015)

Education

PhD, 2021 Department of Astronomy, The University of Tokyo, Japan.
Title: Discovery and Characterization of Transiting Exoplanets with Diverse Radii and Ages (online version). Advisor: Prof. Motohide Tamura

MSc, 2018 Department of Astronomy, The University of Tokyo, Japan.
Title: Multi-color Simultaneous Transit Observations of Low Density Hot Jupiters (online version).
Advisor: Prof. Motohide Tamura

BSc, 2013
Department of Physics, University of the Philippines, Los Baños, Laguna, Philippines.
Title: Thermal Differential and Image Analysis of a Fabricated 11-cm Solar Telescope using an Active-Passive Cooling System. Advisor: Dr. Rogel Mari Sese

Awards

2022/12: St. Joseph's Academy, Centennial Achiever's Award
2021/3: Philippine Embassy in Japan, Medal for distinguished Filipino Graduates
2018/4-2021/3: Ministry of Education, Culture, Sports (MEXT) PhD Scholarship (JPY 8.4M)
2016/4-2018/3: MEXT MSc Scholarship (JPY 5.6M)
2015/4-2016/3: MEXT Japanese Language and Research Scholarship (JPY 2.8M)
2015/10-2016/3: Research Grant from the Graduate University for Advanced Studies (JPY 0.2M)
2014/9: SanDisk Japan Scholarship (USD 3.75k, deferred)
2013/3: International Space University Scholarship (EUR 9k, deferred)
2011-Present: Received multiple travel grants to attend workshops in the US and Taiwan related to data science, and participated in research-related conferences and observation runs in the U.S., Europe, South Africa, and South-east Asia

Teaching and Mentorships

Designed and Taught Courses: Created and delivered two elective courses for 4th-year undergraduate Aerospace Engineering students at the Ateneo de Davao University during the 2nd semester of 2021.

Research Collaborations: Collaborated with M. Mori (Current postdoc) on the discovery and validation of a rare planet in the Neptune desert (Refer to Mori et al. 2022).

Pipeline Development: Wrote a public Python code for easily analyzing data from TESS telescope appropriate for research and instruction (Refer to github.com/jpdeleon/quicklook)

Thesis Advisory: Worked with H. Kobayashi (Former undergraduate student) to develop a pipeline for finding young transiting planets and binary stars in nearby star clusters (Refer to github.com/hiremasa/ytps). Also advised A. Nodado (Former student) on her undergraduate thesis focused on Mira variables.

MEXT Scholarship Guidance: Provided guidance to A. Javier and C. Cabatlao from their scholarship application to acceptance, supporting their successful pursuit of graduate studies in Japan.

Trainings and Workshops

Nature Masterclasses: Impressive Academic Presentations (Expert trainer-led full-day workshop), University of Tokyo, March 18, 2026

University of Tokyo Faculty Development (UTokyo FD Open) Program, Tokyo, Japan, Oct 2025–Feb 2026

5th ExoClock Meeting, Madrid, Spain, Oct 4–6, 2025

ARIEL Consortium Meeting, Madrid, Spain, Sep 30–Oct 4, 2025

Young Transiting Planet Workshop 2025, Ishigaki, Japan, Mar 12–15, 2025

JWST Data Analysis Workshop, Chiang Mai, Thailand, Jun 24–Jul 5, 2024

Japan Planetary Atmosphere Research Society Workshop II, Atami, Japan, Mar 2023

Japan Planetary Atmosphere Research Society Workshop, Matsushima, Miyagi, Japan, Mar 2022

Outreach Activities

Resource speaker for Junior High School Career Week, SJASQ, Philippines, Jan 24, 2026

Guest on a podcast titled "Are We Alone?" (published on Spotify), Dec 2025

Resource speaker for Philippine Astronomy Week 2024 led by PAGASA, Philippines, Feb 2024

Guest at Pinoy Scientists "Pathways to Science" (Aug 2023)

Featured by Pinoy Scientists (Jan 2023)

Resource speaker for Science Week at De La Salle Integrated School, Philippines, Mar 2022

Resource speaker for the Philippine Science High School–Western Visayas Campus, Philippines, Mar 2021

Courtesy call with the Director of the Philippine Space Agency (PhilSA), Mar 2021

Invited speaker at the National Institute of Physics, University of the Philippines Diliman (UPD), Jan 2020

Affiliations and Services

Monthly Notices of the Royal Astronomical Society (MNRAS; Impact Factor: 4.8; active referee)

SPACETIDE Foundation (spacetide.jp; active intern, 2022–present)

Science Integrated Direction for High School Investigators (SIDHI; mentor, 2022–present)

Science and Technology Advisory Council–Japan Chapter (STAC-J; member, 2019–present)

Exoplanet Follow-up Observing Program (ExoFOP; member, 2018–present)